

Exercise 10

Parallel and Distributed Systems

Sockets

In this exercise, we will implement a monitoring chain using TCP sockets and other design options.

Figure 1 shows an overview of our monitoring system. We will have multiple sensors (e.g., temperature and humidity) sending data to the monitoring station. The monitoring station will forward all data to a display.

Task 1 – Temperature Sensor

Implement a Python program that sends a message like *Temperature: 23.3°C* every second via TCP to the monitoring station. Only use one TCP connection. The monitoring station will run at the address 127.0.0.1 at port 5001. The temperature value should be a random number between 18 and 30.

Task 2 – Humidity Sensor

Implement a Python program that sends a message like *Humidity: 56.2%* every 2 seconds via TCP to the monitoring station. Only use one TCP connection. The monitoring station will run at the address 127.0.0.1 at port 5001. The humidity value should be a random number between 0 and 100.

Task 3 – Monitoring Station

Implement a Python program that allows to establish connections and receive TCP data from multiple clients (here: different sensors) and simply forwards the data to a display, also via TCP. Create a new TCP connection to the display for every data you send. Assume the display runs at the address 127.0.0.1 at port 5002.

Task 4 – Display

Implement a Python program that accepts one TCP connection at a time, receives incoming data, and prints it to the console.

Task 5 – Limitations and Outlook

- a) Assume your colleague suggests replacing TCP with UDP here. Collect arguments for and against using UDP in this scenario.

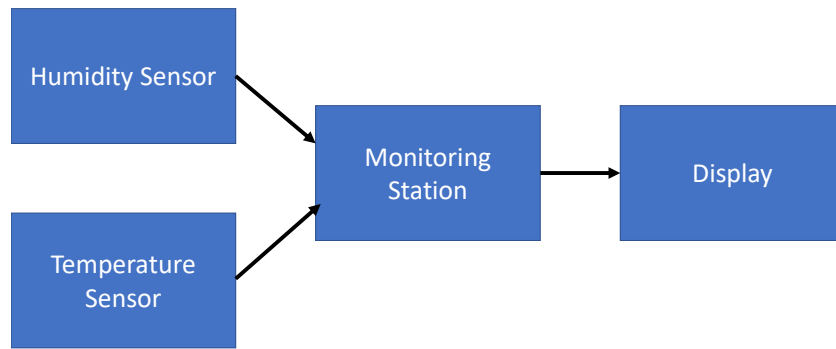


FIGURE 1: System overview.

- b) Explain the performance bottlenecks in the current implementation.

- c) Assume you were not forced to use TCP/UDP and all components (sensors, monitoring station, display) run as processes on one machine. Which IPC mechanism would you choose? Briefly explain your decision.

- d) Assume that we have hundreds of sensors and many displays that only show certain types of data (e.g., one display for temperature only). How would you design the system in this case?